

SIMS CAFE MANAGEMENT SYSTEM

Complete Master Manual: Architecture, Full Library Catalog, Functions, Installation Prerequisites & Teaching Guide

Deployment Targets: Windows Desktop (x64 EXE & Installer) • Android Mobile & POS Tablets (APK)

1. Executive Summary & Project Purpose

SIMS Cafe is a production-grade, enterprise Point-of-Sale (POS) and Restaurant Management ERP ecosystem. Engineered with **Flutter / Dart** for responsive multi-platform interfaces and paired with a high-performance **.NET 8 C# companion microservice (CafePrinter.exe)** for native Win32 printer spooling, SIMS Cafe eliminates the fatal dependency on active internet connections in hospitality venues. The system delivers **sub-second real-time peer-to-peer LAN synchronization** over Wi-Fi via embedded Shelf HTTP and WebSocket protocols, dual-order numbering, multi-station Kitchen Order Ticket (KOT) dispatching, split tender checkout, customer credit ledgers (Khata), and expense reconciliation.

2. Complete Technology Stack & Detailed Library Catalog

Below is the complete, categorized breakdown of all programming languages, frameworks, Flutter packages, and .NET libraries used in SIMS Cafe along with their exact functional role:

A. Core Framework, State Management & Database Libraries

Library / Dependency	Version	Specific Role & Use in SIMS Cafe
flutter & dart	v3.29 / 3.7+	Core cross-platform UI rendering engine, reactive widget tree, event loop, and Dart runtime on Windows and Android.
provider	^6.1.2	App-wide state management via ChangeNotifier. Powers 10 decoupled state managers: OrderProvider, MenuProvider, TableProvider, LanSyncProvider, SettingsProvider, etc.
sqflite	^2.3.3	Native SQLite database driver for Android mobile and POS tablets, managing local CRUD operations and transactions.
sqflite_common_ffi	^2.3.0	C-FFI SQLite loader for Windows Desktop (cafeapp.exe), enabling high-speed local database operations on PC.
shared_preferences	^2.3.3	Key-value persistence for app configuration: printer IP/ports, server host/client mode toggle, store profile, and UI preferences.
flutter_secure_storage	^9.0.0	Encrypted storage for sensitive secrets: admin master PIN, API keys, and device authorization tokens.
path_provider & path	^2.1.4 / ^1.9.0	Resolves OS-specific stable storage paths (AppData/Roaming on Windows, app sandbox on Android) to ensure data persists across app updates.

B. Real-Time Networking, LAN Sync & Cloud Libraries

Library / Dependency	Version	Specific Role & Use in SIMS Cafe
shelf & shelf_router	^1.4.1 / ^1.1.4	Embedded HTTP microserver running inside the host application on Port 8642. Serves REST endpoints for node handshake, full sync, and incremental sync.
shelf_web_socket & web_socket_channel	^2.0.1 / ^3.0.1	WebSocket communication pipeline for sub-second real-time event broadcasting (e.g. ORDER_CREATED, TABLE_OCCUPIED) to all connected waiter tablets.
http & connectivity_plus	^1.3.0 / ^6.0.3	HTTP client for cloud API communication and live network state listener detecting Wi-Fi/Ethernet disconnects and triggering auto-reconnection.
network_info_plus	^5.0.3	Extracts local Wi-Fi IP address (e.g. 192.168.1.100) to broadcast server location to peer devices during auto-discovery.
firebase_core & firestore	^3.4.1 / ^5.0.0	Cloud Firestore for online store registration, license validity verification, and remote telemetry.
firedart	^0.9.8	Pure Dart Firestore implementation used as fallback for Windows desktop without requiring native C++ Firebase runners.
googleapis & google_sign_in	^14.0.0 / ^6.2.1	Google Drive API client allowing automated scheduled and manual encrypted SQLite database zip backups directly to owner's Google Drive.

C. Thermal Printing, Hardware & Graphics Libraries

Library / Dependency	Version	Specific Role & Use in SIMS Cafe
esc_pos_utils_plus & esc_pos_printer_plus	^2.0.0 / ^0.1.1	ESC/POS protocol byte generators: formats receipt layouts, bold text, column alignment, QR codes, paper cuts, and cash drawer kick pulses.
flutter_usb_printer	^0.1.0+1	Direct USB communication for Android POS terminals communicating with thermal printers via USB OTG Vendor/Product IDs.
SkiaSharp & SkiaSharp.HarfBuzz (.NET 8)	2.88.8	High-performance 2D graphics engine in CafePrinter.exe. Performs text shaping with HarfBuzz and rasterizes pixel-perfect 1-bit monochrome Arabic receipts using Cairo/Amiri fonts.
image	^4.2.0	Bitmap image decoding, resizing, color space conversion, and dithering for logo printing on thermal receipts.
pdf & flutter_pdfview	^3.11.3 / ^1.3.2	Vector PDF document generator and in-app interactive PDF viewer for sales reports, tax audits, and catering quotation sheets.
excel	^4.0.3	Spreadsheet parsing and generation engine. Enables bulk Excel menu catalog imports and end-of-day sales data workbook exports.
printing	^5.14.2	Standard OS document printing integration for standard A4/Letter invoice and report printouts.

D. Desktop OS Integration, Media & Utility Libraries

Library / Dependency	Version	Specific Role & Use in SIMS Cafe
window_manager	^0.4.2	Controls native Windows desktop window behavior: minimum dimensions (1024x768), kiosk fullscreen mode, and custom title bars.
file_picker & desktop_drop	^8.1.3 / ^0.4.4	Native file picker dialogs and drag-and-drop support on Windows, allowing cashiers to drag Excel menu files directly into the app.
camera & camera_windows	^0.10.5 / ^0.2.1	Accesses webcam on Windows and camera hardware on Android for barcode scanning and taking product photos.
image_cropper & crop_your_image	^11.0.0 / ^2.0.0	In-app interactive image cropping and framing widget for menu item photos and company logo uploads.
audioplayers	^6.6.0	Low-latency audio playback for auditory feedback (e.g. order placed sound, KOT alert chime, error alert).
auto_updater & upgrader	^0.1.7 / ^10.3.0	Automated background update engines: Appcast XML feed updater for Windows EXE and in-app update prompt for Android APK.
intl & flutter_dotenv	^0.19.0 / ^5.2.1	Internationalization, date/currency formatting (DateFormat, NumberFormat), and runtime .env file configuration loader.
msix	^3.16.6	Windows MSIX package builder generating digitally signed Windows Store and enterprise deployment bundles.

3. Functions & Features Catalog

SIMS Cafe incorporates a full suite of restaurant and retail automation capabilities organized into functional modules:

Module	Function / Feature	Detailed Technical & Business Capabilities
1. Multi-Service POS & Ordering	Dine-In, Takeout, Delivery, Drive-Thru, Catering	• Dine-In Table Grid: Real-time floor plan with visual occupancy (Free, Occupied, Billed), split bill, and table merge. • Takeout / Fast Food: Instant order entry with automated queue token generation. • Home Delivery: Delivery boy assignment, customer address/contact tracking, and delivery surcharge. • Drive-Through: Quick vehicle lane queueing with license plate recording. • Catering Bookings: Event date/time, guest count, advance deposit logging, and balance tracking.
2. Split Tender & Checkout	Flexible Payment Matrix & Invoicing	• Split Payments: Concurrently accept Cash, Card/Bank, Customer Credit, and Advance Deposit on a single invoice. • Taxation & Discounts: Configurable VAT computation, item-level tax exemptions, and fixed or percentage discounts. • Change Calculator: Dynamic change-due calculator with one-tap quick cash denomination buttons. • Temporary Bill Preview: Print preliminary guest check receipts before final payment settlement.
3. Kitchen Dispatch (Multi-KOT)	Intelligent Station Routing	• Multi-Printer Matrix: Splits an order across multiple kitchen stations (e.g. Barista, Hot Kitchen, Dessert station). • Modifier Notes: Specific preparation notes per item (e.g. 'extra ice', 'no onion', 'well done'). • Duplicate Safety: Tracks printed items to prevent duplicate tickets on order updates.
4. Menu & Inventory	Product Catalog Management	• Multi-Size Pricing: Assign size variations (Small, Medium, Large, Full, Half) with dynamic pricing. • Barcode Scanner: Scan physical product barcodes for retail cafe snacks and beverages. • Profit Margin Tracking: Records purchase cost vs selling price to compute live gross profit. • Excel Bulk Import/Export: Add hundreds of menu items in seconds via formatted Excel spreadsheets.
5. Customer Khata & Credit	Customer Ledger & Debt Accounting	• Customer Directory: Searchable customer CRM with contact numbers, addresses, and transaction history. • Credit Sale Tracking: Bill orders directly to customer credit accounts. • Repayment Ledger: Partial or full debt settlements with printed thermal payment vouchers.
6. Expense & Cash Ledger	Daily Overhead Reconciliation	• Daily Expense Logging: Record vendor payouts, raw ingredient purchases, utility bills, and petty cash. • Cashier Shift Closing: Automatically balances opening float + cash sales - cash expenses = drawer balance.
7. Business Intelligence	Reports & Export Analytics	• Interactive Dashboard: Real-time charts for revenue growth, order volume, and hourly peak times. • Item-Wise Sales & Top Sellers: Identifies top-performing dishes and slow-moving inventory. • Export Formats: One-click export to PDF report summaries and structured Excel workbooks.
8. Security & Recovery	Role Guards & Crash Protection	• PIN-Protected Admin Functions: Master PIN lock on Settings, Discount overrides, and Financial Reports. • Safe Mode Detector: Dynamic fallback mode bypassing window/graphics crashes on damaged Windows setups. • Cloud Google Drive Backup: Scheduled or manual zip archive database backup and restore.

4. System & Installation Requirements

To deploy, run, or build SIMS Cafe, the following hardware, operating system, and software specifications are required:

A. Hardware & Operational Specifications

Component	Host PC / Main Cashier Terminal	Client Device (Waiter Tablet / Mobile)
Processor (CPU)	Intel Core i3 / AMD Ryzen 3 (or higher, x64 architecture)	Quad-core 1.8 GHz ARM processor (e.g. Snapdragon, MediaTek)
Memory (RAM)	Minimum 4 GB RAM (8 GB recommended for heavy traffic)	Minimum 2 GB RAM (3 GB+ recommended for smooth UI)
Disk Storage	Minimum 500 MB free space (SSD recommended for SQLite I/O)	Minimum 150 MB free internal flash storage
Display Resolution	1366x768 or 1920x1080 (Touchscreen monitor supported)	7.0" to 11.0" Tablet display or 5.5"+ Mobile Phone screen
Network (Wi-Fi)	Wi-Fi Router / Local LAN Switch (Same subnet, e.g. 192.168.1.x)	Connected to the same local Wi-Fi network as the Host PC
Thermal Printer	80mm or 58mm ESC/POS Thermal Receipt & KOT Printer (USB, LAN, or COM)	Network TCP/IP Thermal Printer or USB OTG Thermal Printer

B. Operating System & Runtime Prerequisites

- **Windows Desktop:** Windows 10 (Build 19041+) or Windows 11 (64-bit). Requires *Visual C++ 2015-2022 Redistributable*.
- **Android Environment:** Android 6.0 (API Level 23) up to Android 15. Requires Camera and Local Storage permissions.
- **Local Network Rules:** Windows Firewall must allow inbound traffic on TCP Port **8642** (LAN sync) and Port **9100** (Raw printer socket).

C. Developer & Build Toolchain Requirements

If compiling the project from source code, the following tools must be installed on the build workstation:

Development Tool	Required Version	Purpose in Build Process
Flutter SDK	v3.29.0 or higher (Channel stable)	Compiles Flutter Dart application for Windows x64 and Android APK.
.NET SDK	.NET 8.0 SDK (x64)	Compiles CafePrinter.csproj into self-contained single-file CafePrinter.exe.
Inno Setup	Inno Setup 6.x (ISCC.exe)	Packages Flutter binaries, C# printing microservice, assets, and icons into setup wizard.
Visual Studio	VS 2022 (Desktop C++ workload)	Provides MSVC compiler and CMake toolchain for Flutter Windows desktop runner.
Android SDK / Java	JDK 17 + Android SDK API 34	Compiles and signs release APK packages with Gradle 8+.

D. Step-by-Step Installation Guides

1. Installing on Windows Desktop (End-User):

- Step 1: Run `SimsCafe_Setup_v2.0.exe`.
- Step 2: Follow the setup wizard to install into `C:\Program Files\SIMS CAFE`.
- Step 3: Launch SIMS CAFE from Desktop shortcut. The app automatically creates its databases in `%APPDATA%\com.example.cafeapp/databases/`.
- Step 4: Open *Printer Settings* to select your receipt printer and set Host Mode to **Server**.

2. Installing on Android Tablets/Mobiles (End-User):

- Step 1: Transfer `app-release.apk` to the device via USB, WhatsApp, or local download link.
- Step 2: Enable 'Install from Unknown Sources' and tap the APK to install.
- Step 3: Connect the tablet to the cafe's Wi-Fi network. Open SIMS CAFE -> *Device Management*.
- Step 4: Tap **Auto-Discover Host** or enter Host IP (e.g. `192.168.1.100:8642`) to sync all menu and table data.

5. Local Database Architecture & SQLite Schemas

The system isolates its storage into **six dedicated SQLite databases** located in the persistent OS application directory. Each database operates with Write-Ahead Logging (WAL) and explicit version migration scripts.

A. `cafe_orders.db` — Orders & Items Schema

Field Name	Data Type	Constraints & Functional Role
<code>id</code>	INTEGER	Primary Key, AUTOINCREMENT
<code>staff_order_number</code>	INTEGER	Staff device local order counter (prevents race conditions)
<code>main_order_number</code>	INTEGER	Synchronized master order sequence assigned by server
<code>staff_device_id</code>	TEXT	Originating device identifier / UUID
<code>service_type</code>	TEXT	Dining (Table X), Takeout, Delivery, Drive Through, Catering
<code>subtotal, tax, discount</code>	REAL	Financial components calculated before final net total
<code>total</code>	REAL	Net payable amount after applying tax and discounts
<code>payment_method</code>	TEXT	cash, bank, credit, deposit, split
<code>cash_amount, bank_amount</code>	REAL	Individual split payment contributions
<code>customer_id, customer_name</code>	TEXT	Linked customer record for Khata / credit billing
<code>delivery_boy, delivery_address</code>	TEXT	Delivery dispatch information and rider assignment
<code>event_date, event_guest_count</code>	TEXT / INT	Catering specific booking metadata
<code>deposit_amount</code>	REAL	Advance deposit paid for catering bookings
<code>token_number</code>	TEXT	Customer queue token printed for takeout service
<code>status</code>	TEXT	pending, completed, cancelled
<code>created_at, updated_at</code>	TEXT	ISO 8601 timestamps for Last-Write-Wins conflict resolution
<code>is_synced, is_deleted</code>	INTEGER	LAN sync status flag (0/1) and soft-deletion tombstone

B. `order_items` (Child Table with CASCADE Delete)

Field Name	Data Type	Description & Relationship
<code>id</code>	INTEGER	Primary Key, AUTOINCREMENT
<code>order_id</code>	INTEGER	Foreign Key -> <code>orders(id)</code> ON DELETE CASCADE
<code>menu_item_id</code>	INTEGER	Referenced menu product ID
<code>name, price, quantity</code>	TEXT/REAL/INT	Snapshot of item details at time of order creation
<code>kitchen_note</code>	TEXT	Custom cooking instructions (e.g. 'extra spicy', 'no sugar')
<code>tax_exempt, purchase_price</code>	INT / REAL	Tax zero-rating flag and purchase cost for gross profit reports

C. Supporting Databases Overview

Database File	Primary Table	Key Columns & Purpose
cafe_menu.db	menu_items	id, name, price, imageUrl, category, isAvailable, isDeleted, lastUpdated, taxExempt, isPerPlate, purchasePrice, barcode, sizes (JSON).
cafe_persons.db	persons	id, name, phoneNumber, place, dateVisited, credit, updated_at, is_deleted. Stores customer CRM profile and cumulative credit debt.
credit_transactions.db	credit_transactions	id, customerId, customerName, orderNumber, amount, createdAt, serviceType, isCompleted, updated_at. Double-entry audit ledger for credit sales & repayments.
cafe_expenses.db	expenses & expense_items	id, date, cashier, accountType, grandTotal, createdAt + items: sINo, account, narration, amount, remarks. Tracks overheads and vendor bills.
cafe_delivery_boys_store.db	delivery_boys	id, name, phoneNumber, updated_at, is_deleted. Roster of delivery drivers assigned to takeout deliveries.

6. Networking, APIs & Hardware Integration

A. Embedded Host REST API Endpoints (Shelf Server :8642)

Endpoint	Method	Payload & Description
/api/ping	GET	Heartbeat & server discovery. Returns server name, IP, and timestamp.
/api/sync/full	GET	Transmits complete snapshot of Orders, Menu, Customers, and Tables for new node onboarding.
/api/sync/incremental	POST	Differential sync: Client sends {since: timestamp}, server returns only updated records.
/api/sync/push	POST	Client pushes newly created orders and customer edits directly to host database.
/ws	WS	Bi-directional real-time WebSocket channel broadcasting ORDER_CREATED, TABLE_OCCUPIED, etc.

B. Thermal Printing & Hardware Control

Printer Mode	Target Environment	Implementation & Capabilities
Windows Spooler (RAW)	Windows Desktop ('cafeapp.exe')	Delegates to CafePrinter.exe (.NET 8). Uses Win32 winspool.drv APIs (OpenPrinter, StartDocPrinter, WritePrinter) to send raw ESC/POS binary directly to USB/LAN printers.
Network TCP Socket	Android Tablet / Windows	Direct raw TCP socket connection on Port 9100 . Connects directly to Ethernet/Wi-Fi thermal printers with 3-second timeout protection.
Android USB Direct	Android POS Hardware	Uses flutter_usb_printer to communicate directly with thermal printers via USB OTG Vendor ID / Product ID endpoints.
Arabic RTL Rasterizer	Multilingual Receipts	Employs SkiaSharp bitmap rasterization with Amiri and Cairo fonts. Generates pixel-perfect 1-bit monochrome bitmaps for ESC/POS printers lacking Arabic hardware code pages.

7. Build & Compilation Guide (Windows EXE & Android APK)

A. Automated Windows Build Workflow

Windows deployment is automated via cafeapp/build_installer.bat:

1. Publish C# Printing Service:

```
cd CafePrinter
dotnet publish -c Release -r win-x64 --self-contained -p:PublishSingleFile=true -o publish\
```

2. Build Flutter Windows Release:

```
cd ..\cafeapp
flutter clean && flutter build windows --release
```

3. Copy Companion Executable:

Copy CafePrinter.exe into build\windows\x64\runner\Release\.

4. Compile Inno Setup Installer:

```
"C:\Program Files (x86)\Inno Setup 6\ISCC.exe" sims_cafe_installer.iss
Output: installer_output/SimsCafe_Setup_v2.0.exe
```

B. Android Release APK Compilation

1. Fetch Dependencies: flutter pub get

2. Build Release APK: flutter build apk --release --split-per-abi

3. Output File: build/app/outputs/flutter-apk/app-release.apk

8. Presentation & Teaching Manual (Demo Script & Viva Q&A;)

A. 2-Minute Elevator Pitch

"SIMS Cafe is an enterprise-grade, offline-first Restaurant POS and ERP ecosystem built with Flutter and .NET. Unlike traditional cloud POS systems that halt when internet connections drop, SIMS Cafe runs 100% locally with high-performance SQLite databases, while maintaining real-time sub-second sync across all cashier terminals and waiter tablets over local Wi-Fi. It handles end-to-end cafe operations: dynamic table management, multi-station kitchen ticket routing, split payments, customer credit tracking, VAT accounting, expense ledgers, and automated cloud backups."

B. Step-by-Step Live Demonstration Script

- Host Launch & Discovery:** Launch the Windows desktop app. Show that the embedded Shelf server starts immediately on port 8642. Connect an Android tablet or second window to demonstrate auto-discovery.
- Dining Table Order:** Open Table Grid. Select Table 4. Add 2 Cappuccinos + 1 Burger with 'extra sauce'. Show that Table 4 instantly turns orange (occupied) on all other connected devices in real time via WebSockets.
- Multi-KOT Kitchen Routing:** Tap 'Send to Kitchen'. Explain how the system splits the items—sending the burger to the kitchen printer and the coffee to the barista printer.
- Tender & Split Checkout:** Go to checkout. Split payment (\$10 cash + \$15 card) and demonstrate VAT/discount calculation. Complete the order to trigger the final ESC/POS customer receipt with QR/Barcode.
- Customer Credit (Khata):** Demonstrate selecting a registered customer, charging an order to credit, and opening Customer Management to show the updated debt ledger and repayment settlement.
- Day-End Analytics & Export:** Open Reports screen to show live revenue charts, tax breakdown, and item sales. Export a clean PDF/Excel report with one click.

C. Technical Viva / Defense Questions & Model Answers

Question 1: Why did you choose SQLite over a single centralized cloud database (like MongoDB or MySQL)?
Answer: Hospitality businesses cannot tolerate downtime. If the internet drops during peak dining hours, a cloud-dependent POS freezes, causing severe revenue loss. SQLite provides zero-latency, ACID-compliant local storage on every device. Our embedded Shelf LAN sync layer bridges devices locally, offering cloud-like synchronization without internet dependency.
Question 2: How do you prevent duplicate order numbers when multiple waiter tablets take orders simultaneously offline?
Answer: We implemented a <i>Dual Numbering Architecture</i> . Each staff device maintains its own local counter (staff_order_number). When the order reaches the host or is completed, the master terminal assigns the official synchronized main_order_number. This completely eliminates race conditions.
Question 3: Why is there a separate C# project (CafePrinter.exe) instead of using Flutter printing plugins directly?
Answer: Flutter desktop printing plugins often struggle with raw ESC/POS byte streaming, device status interrogation (detecting paper-out/offline queues), and Arabic typography shaping. CafePrinter.exe leverages Win32 winspool.drv and SkiaSharp to provide native spooler access, instant offline detection, and pixel-perfect Arabic rendering.
Question 4: How does the system handle database migrations when releasing app updates on Windows?
Answer: In DatabaseHelper.dart, we store databases in a stable OS-managed directory (AppData/Roaming/com.example.cafeapp/databases/). When updating, the app auto-detects legacy database files, copies WAL/SHM journal files safely, and executes SQLite onUpgrade scripts incrementally.